

INITIAL PLANNING

Project: Employee Management System – A Square Skills Academy

1. Project Summary

The **Employee Management System** is a web-based application for A Square Skills Academy that streamlines employee management, task assignment, attendance tracking, and internal communication between admins and employees. It allows admins to manage employees, assign and monitor tasks, and generate reports, while employees can view tasks, submit updates, track attendance, and communicate efficiently. The goal is to enhance organizational productivity and transparency.

2. Technologies Usage

- **Backend:** Python (FastAPI)
 - **Database:** MongoDB (Local Instance)
 - **Frontend:** HTML, CSS, JavaScript
 - **Design Tool:** Figma (UI/UX Design)
 - **Development Environment:** Visual Studio Code
 - **Version Control:** Git & GitHub (for code management)
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3. Procedure

1. **Requirement Analysis**
 - Gather all functional and non-functional requirements.
 - Identify user roles (Admin, Employee) and their permissions.
 - Define database structure and data flow.
2. **Planning**
 - Create detailed workflow diagrams and site map.
 - Divide work modules among team members.
 - Schedule milestones and deadlines.
3. **Designing**
 - Develop wireframes and final UI designs on **Figma**.
 - Focus on user experience, layout consistency, and color themes.
 - Prepare responsive and modern component designs.
4. **Development (Coding Phase)**
 - Use **Visual Studio Code** as the main IDE.
 - Backend with **FastAPI** and MongoDB for CRUD operations.
 - Frontend with **HTML, CSS, JS** for a clean and responsive interface.
 - Integrate backend APIs with the frontend.

5. Testing & Deployment

- Test all functionalities (unit, integration, and UI testing).
 - Fix bugs and optimize performance.
 - Deploy locally or on an internal server.
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4. Modules and Authority

Admin

- Create, Read, Update, Delete Employees.
- Assign, edit, and track tasks (with attachments and deadlines).
- View and approve leaves.
- Track employee attendance and login/logout records.
- Generate performance reports (monthly, quarterly, yearly).
- Promote employees to admin.
- Communicate with employees via internal message portal.

Employee

- View and comment on assigned tasks.
 - Mark tasks complete/incomplete (with undo option).
 - Download attachments from assigned tasks.
 - View their own login/logout sessions and work hours.
 - Communicate with admins via message portal.
 - Apply for and track leave requests.
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5. Site Map

Authentication

- Login Page
- Forgot Password / Reset Password
- Error Page (404)

Admin Dashboard

- Dashboard Overview
- Employee Management
 - All Employees
 - Create / Edit / Delete Employee
 - Employee Detail View
- Task Management
 - All Tasks
 - Create / Edit / View Task

- Reassign Task
- Attendance & Reports
 - Daily Attendance Logs
 - Employee Performance Report
 - Export Reports (PDF/CSV)
- Leave Management
 - All Leave Requests
 - Leave Details (Approve/Reject)
- Messaging & Notifications
 - Message Portal (Admin ↔ Employee)
 - Notifications Panel
- Settings
 - Profile Settings
 - Role Management (Promote Employee)

Employee Dashboard

- Dashboard Overview
- My Tasks
 - Task List
 - Task Detail (with Comments & Attachments)
- Communication
 - Message Portal
 - Notifications
- Attendance & Reports
 - My Attendance Log
 - My Performance Report
- Leave Management
 - Apply for Leave
 - My Leave Requests
- Profile & Settings
 - My Profile
 - Change Password